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[Total No. of Questions - 9] [Total No. of Printed Pages : 2]

May-26-0603

MCA-6206(iii) (Information Security)

MCA 2nd (NEP)

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all selecting at least one question each from of section A, B, C & D and all the subparts of the questions in section E.

SECTION - A

1. Describe the Security Architecture for Open Systems. How does it help to ensure interoperability and security across different systems? (12)
2. Differentiate between symmetric key encryption and asymmetric key encryption. Describe their advantages and limitations. (12)

SECTION - B

3. (a) What is password-based authentication? Describe its strengths and weaknesses. (6)
(b) Define subjects, objects, and access rights in the context of access control. (6)
4. Explain the working of UNIX File Access Control. How do permissions like 'rwx' work for owner, group, and others? (12)

SECTION - C

5. (a) Explain the security challenges faced by Database Management Systems (DBMS). (6)

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- (b) What are the common defense mechanisms against DoS/DDoS attacks? (6)
6. (a) What is database access control? Explain the different levels at which access control can be implemented in a database system. (6)
(b) What is database encryption? Explain its role in protecting sensitive information stored in databases. (6)

SECTION - D

7. (a) Describe five major physical security threats that can compromise the confidentiality and integrity of any organization. (6)
(b) Define physical security. What are its key components? (6)
8. Identify and explain different types of insider threats in physical security. What are the different ways to mitigate them? (12)

SECTION - E (Compulsory)

9. (a) What are the main categories of security threats?
(b) Define user authentication.
(c) Define Phishing.
(d) What are bandwidth attacks?
(e) Describe the importance of physical and infrastructure security.
(f) How can high temperature affect IT infrastructure in an organization? (6×2=12)